

SOC INCIDENT INVESTIGATION REPORT

1. Executive Summary

On **1 February 2026**, the Security Operations Center (SOC) identified multiple failed SSH login attempts targeting a Linux server. The abnormal authentication behavior was detected through log analysis of `/var/log/auth.log` using a custom Python automation script. Investigation confirmed an SSH brute-force attack originating from an external IP address. No successful privilege escalation or data compromise was observed. The incident was contained by blocking the attacker IP and hardening SSH configurations.

2. Tools and Frameworks Used

Category	Tool	Purpose
Log Source	Linux auth.log	SSH authentication events
Automation	Python	Log parsing & detection
OS	Ubuntu Linux	Victim system
Threat Intel	AbuseIPDB	IP reputation check
Framework	MITRE ATT&CK	Attack classification

3. Incident Description

3.1 Detection Summary

- **Detection Method:** Python SSH log analyzer
- **Attack Type:** SSH Brute Force
 - Log File:** `/var/log/auth.log`
- **Event Count:** Multiple failed login attempts

Target Service: SSH

Timeframe: 1 February 2026 (08:00 UTC – 10:30 UTC)

3.2 Event Details

Failed password for invalid user root from 192.168.56.104 ssh2

Failed password for invalid user admin from 192.168.56.104 ssh2

4 Investigation Process

Step 1: Log Collection Authentication logs were collected from /var/log/auth.log, which records all SSH login attempts on Linux systems.

Step 2: Automated Analysis A Python script was executed to extract source IP addresses, count failed login attempts, and identify IPs exceeding the defined threshold.

Step 3: Threat Intelligence Validation The attacker IP was checked on AbuseIPDB, confirming previous reports of SSH brute-force activity.

Step 4: MITRE ATT&CK Mapping

Tactic	Technique ID	Technique Name
Credential Access	T1110	Brute Force
Initial Access	T1078	Valid Accounts

Step 5: Correlation and Verification

Findings:

- **Attack Type:** SSH brute-force

- **Source:** External malicious IP
- **Successful Login:** No
- **Privilege Escalation:** Not detected

Impact Assessment:

- **System Impact:** Low (No compromise)
 - **Business Impact:** Low (Attempt blocked)
 - **Risk Level:** Medium (Persistent attempts)
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5 Findings and Impact Assessment

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6 Mitigation and Recommendations

- **Block attacker IP** using iptables or ufw firewall rules.

- **Disable password-based SSH login** in `/etc/ssh/sshd_config`.
 - **Enable SSH key-based authentication** for all users.
 - **Change default SSH port** from 22 to a non-standard port.
 - **Implement Fail2Ban** to automatically jail IPs with repeated failures.
 - **Continuous log monitoring** and real-time alerting.
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7 Conclusion

The SOC successfully detected and analyzed an SSH brute-force attack using automated log analysis. Early detection prevented unauthorized access. Implementing stronger SSH security controls will further reduce future attack risks.

Final Status:  **Incident Contained – No System Compromise**

8 Appendix

- **Log Source:** `/var/log/auth.log`
- **Detection Script:** `ssh_bruteforce_detector.py`
- **Reference:** MITRE ATT&CK v14 – T1110 (Brute Force)